

GRAVIMETRIC ANALYSIS NOTES

Gravimetric analysis is a quantitative analytical technique that involves the determination of the amount of an analyte present in a sample by measuring the mass of a product that is formed or a reactant that is lost. The principle upon which gravimetric analysis is based is that the mass of a substance can be accurately determined by weighing it, and by applying stoichiometric relationships, the mass of the analyte in a sample can be calculated.

Precipitation Gravimetry

This is a gravimetric analysis technique that involves the separation of an analyte from a sample solution by the addition of a precipitating reagent. The precipitate formed is then collected, washed, and dried before being weighed. The weight of the precipitate is used to determine the amount of the analyte present in the sample. The key principle of this method is that the analyte and the precipitating reagent react quantitatively to form a product that is insoluble in the solvent.

Volatilization Gravimetry

This is a gravimetric analysis technique that involves the separation of an analyte from a sample solution by the conversion of the analyte into a volatile compound. The volatile compound is then driven off by heating the sample, and the residue left behind is weighed. The weight of the residue is used to determine the amount of the analyte present in the sample. The key principle of this method is that the analyte is converted into a volatile compound that can be driven off by heating, leaving behind a residue of known composition.

Gravimetric Determination of Chlorides

Chlorides can be determined by:

1. Weighing a sample of the solid metal chloride.
2. Dissolving the sample in water.
3. Adding nitric acid then excess silver nitrate to the chloride solution.
4. The chloride precipitates as silver chloride.
5. Collecting the chloride formed by filtration.
6. Washing the chloride with distilled water.
7. Drying the chloride.
8. Weighing the mass of silver chloride formed.

Gravimetric Analysis Problems:

1. In the analysis of 0.7011 g of an impure chloride containing sample, 0.9805 g of AgCl were precipitated. What is the percentage by mass chloride in the sample?
2. A 0.4054 g solid organic sample containing covalently bound bromide and no other halogens was placed in a porcelain crucible with about one gram of fresh sodium metal. In a process known as sodium fusion, the mixture was heated in a furnace to 450 °C which charred and vaporized the organic portion of the molecule and converted the covalent bromide into sodium bromide. Excess sodium was decomposed by adding small portions of water which also dissolved the sodium bromide. The clear solution was quantitatively transferred to a beaker, acidified with dilute nitric acid, and diluted to about 50 mL. A 6 mL quantity of 0.1 M AgNO₃ was added to the solution and the mixture heated to about 60 °C for an hour to age and digest the precipitate. After filtering, the mass of the silver bromide produced was determined to be 37.8 mg. What is the percentage by mass bromine in the organic compound?
3. A 1.1105 g sample of bauxite (the primary ore of aluminium) was analyzed for aluminium. The sample was pulverized and dissolved in concentrated nitric acid. The HNO₃ was removed by evaporation and solids dissolved in hot water with a very small quantity of nitric acid added. Insoluble solids were removed by gravity filtration. The solution was made basic by slow addition of dilute NH₃ at which point a gelatinous solid precipitated, Al (OH)₃ and Al₂O₃.xH₂O. The precipitate was heated to coagulate it as much as possible then filtered over "fast" filter paper. The paper and solids were placed in a porcelain crucible and the paper ashed and precipitate "ignited" at 600 °C to convert the precipitate to pure Al₂O₃. The mass of Al₂O₃ isolated was 0.3605 g. The average mass of ash remaining after charring 10 sheets of the identical filter paper was 0.0006 g. What is the percentage aluminium in the bauxite ore?
4. A 0.8870 g sample containing only NaCl and KCl was treated with AgNO₃. The AgCl formed had a mass of 1.913 g. Calculate the % Na and % K in the sample.